Eric J. Pabon Cancel

Resume

Contact Information

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Education & Academic Background

Ph.D. in Mathematics — Expected June 2029

Purdue University, West Lafayette, Indiana | Advisor: Prof. Aaron Yip

M.S. in Mathematics — Expected May 2026

Purdue University, West Lafayette, Indiana

B.S. in Pure Mathematics, Minor in Applied Math (*Magna Cum Laude*) — December 2022

University of Puerto Rico, Mayagüez Campus (UPRM), Mayagüez, Puerto Rico | Mentor: Reyes M. Ortiz Albino

Skills and Other Information

Programming & Computation: Python, Julia, C++, SageMath, MATLAB | Formatting & Tools: HTML, LaTeX, Git
Math & AI/ML: NumPy, SciPy, PyTorch, TensorFlow, JAX, Flux, CUDA | Spoken Languages: English and Spanish

Research Experience

URA-Sandia Graduate Student Summer Fellow, URA-Sandia National Laboratories — May 2025–August 2025

- Studied machine learning surrogate models, and learned and applied data assimilation for closure models with JAX.
- Conducted a parametric study of the optimization step for 100 noisy samples of an SIQR epidemic model.
- Determined a minimum ensemble size of 500 ensembles for optimal Ensemble Kalman filter-based data assimilation.

GEM Fellow Summer Research Program Intern, MIT Lincoln Laboratory — May 2023–August 2023

- Developed mathematical algorithms for autoencoders with LSTM architecture, using the Flux library.
- Identified the autoencoder that minimized the loss function by using PCA and clustering techniques.
- Used K-medoids to obtain an optimal silhouette score of 0.875, and IF-1 score of 0.839.

Papers and Articles (The asterisk symbol (*) denotes alphabetical order authorship)

[1] S. Polk, E.J. Pabon-Cancel, R. Paleja, K. Chestnut-Chang, R. Jensen and M. Ramírez. Unsupervised Network-Based Behavior Inference from Human Action Sequences (UNBIAS). *2024 IEEE Conference on Games (CoG) 2024, pp. 1-8.*

[2] *D. Chen, P.E. Harris, J. Carlos Martinez Mori, E.J. Pabon-Cancel and G. Sargent. Permutation-Invariant Parking Assortments. *Enumerative Combinatorics and Applications*, **4:1**, 1-25 (2024). #S2R4.

[3] *I. Byrne, N. Dodson, R. Lynch, E.J. Pabon-Cancel and F. Piñero-González. Improving the Minimum Distance Bound of Trace Goppa Codes. *Designs, Codes and Cryptography*. **91**, 2649–2663 (2023).

Projects

Purdue University, West Lafayette, Indiana — January 2025–May 2025

MA59500OT: Computational Optimal Transport and Deep Generative Models

Normalizing Flow Optimal Transport Implementation on the MNIST Dataset

- Developed a normalizing flow neural network that learned the optimal transport path of a Gaussian-distributed MNIST image to the target MNIST number distribution.
- Generated recognizable digit images upon training completion.

WGAN and Monge Map Implementation on the MNIST Dataset

- Constructed a Wasserstein Generative Adversarial Network with Gradient Penalty (WGAN-GP) and Monge Map Network and applied it to the MNIST dataset.
- Successfully generated realistic-looking sample numbers at the end of training and compared with real samples.

Awards

2026 URA STEM Ambassador — March 2026

2025 Universities Research Association-Sandia National Labs Graduate Summer Fellowship — May 2025–August 2025

2023 National GEM Consortium PhD Science Fellowship — August 2023–May 2024

- Purdue University Department of Mathematics Sponsorship — August 2023–May 2024
- MIT Lincoln Laboratory Employer Sponsorship (Internship) — May 2023–August 2023

2023 MIT Lincoln Laboratory I^3C 3rd Place Research Proposal Prize — July 2023

2023 Ford Foundation Predoctoral Fellowship Honorable Mention — March 2023

2022 Evertec Inc. STEM Scholarship — October 2022

Puerto Rico-Louis Stokes Alliance for Minority Participation Research Scholarship — August 2019–December 2022